Retail Profit Analysis

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Abstract

A machine learning model to predict individual retailers' profits based on their previous sales transactions using the Sample Superstore dataset has been developed. In order for retailers to set their optimal pricing strategy, manage their discount policies and allocate their resources effectively, they need to be able to accurately predict their profit based on future sales transactions. A Random Forest Regressor was created using a variety of historical order data features, including sales amount, discount percentage, total quantity sold per order, the product category, shipping mode, region of shipping, shipping duration (which was engineered) and the month of the order (which was also engineered). The performance of the model was assessed through standard regression metrics, achieving an R^2 value of approximately 0.875 (indicating a predictive accuracy of about 87.5%), an MAE of about $14.23 and an RMSE of about $38.71. In terms of the most important predictors of profit based on the Random Forest's feature importance analysis for this dataset, sales amount and discount percentage were the two most significant variables in predicting profit in a retail setting, followed by total quantity sold per order and the product's sub-category. The insights gained from this research provide retailers with vital business information on how to price their products or determine their discount policies.

1. Introduction

Profitability analysis has become a key area of focus in analytics owing to the highly competitive nature of the retail sector. Any enterprise endowed with massive transactional data can utilize machine learning algorithms to unearth insights that lead to or detract from profitability. The 'Superstore' dataset is a fabricated retail database used extensively in data science communities and features transactional data such as items, customer segments, region, discounts offered, and profit made.

The goal of this project is to develop a machine learning algorithm for prediction of profit that a particular future order would generate. Some potential use cases include:

- Identification of products or regions where negative margins are possible
- Indication of orders that might result in losses owing to offered discounts
- Scenario modeling for pricing considerations
- Inventory management

Random forest regressor has been chosen for this exercise because of its effectiveness, flexibility in handling diverse feature types, non-vulnerability to overfitting issues, and built-in capacity to rank features on their importance.

2. Literature Review

Machine learning applications in retail analytics have been studied in depth. Early research on profit and sales forecasting used statistical time-series models like ARIMA and exponential smoothing. While these models are effective for overall trends, they do not make good use of item-level features.

Ensemble tree-based methods, especially Random Forests introduced by Breiman in 2001, have performed well in tabular regression tasks. Their strategy of averaging predictions from many decision trees helps reduce variance without increasing bias. Gradient Boosted Trees, like XGBoost and LightGBM, have become strong alternatives. However, Random Forests still serve as a dependable baseline because they are easier to interpret and less sensitive to hyperparameters.

The Superstore dataset is commonly used in academic and educational settings. Research shows that discount-related features are the main drivers of profit prediction on this dataset, often outweighing geographic and temporal details. This finding reflects well-known insights in retail pricing literature, where deep discounting significantly contributes to margin loss.

Feature engineering, especially extracting temporal signals like shipping duration, day of the week, and seasonal order month, has been proven to improve model accuracy in retail forecasting. This project uses these engineered features to boost predictive power.

3. Methodology

3.1 Dataset

The Sample Superstore dataset was obtained from the Excel file 'sample_-_superstore.xls', specifically from the 'Orders' sheet. The dataset contains 9,994 rows and 21 columns representing individual order line items. Key columns include:

- Order Date and Ship Date (datetime)
- Ship Mode (categorical: Standard Class, Second Class, First Class, Same Day) • Segment (Consumer, Corporate, Home Office)
- Region, State, City (geographic features)
- Category, Sub-Category (product hierarchy)
- Sales, Quantity, Discount (numeric transaction features)

• Profit (target variable - continuous numeric)

3.2 Data Preprocessing

The following preprocessing steps were applied to the raw dataset before model training:

• Date Conversion: Order Date and Ship Date were parsed into Python datetime objects.

• Feature Engineering: Four new features were derived — Order Year, Order Month, Order Day, Order Weekday, and Shipping Days (the difference in days between Ship Date and Order Date).

• Column Dropping: High-cardinality identifier columns (Row ID, Order ID, Customer ID, Customer Name, Product ID, Product Name) and the original date columns were dropped to avoid data leakage and reduce noise.

• Missing Value Check: No null values were found in the dataset, so imputation was not required.

• Encoding: Categorical features were one-hot encoded using pandas get_dummies() to convert them into numeric form suitable for the Random Forest model.

3.3 Model Selection

A Random Forest Regressor from the scikit-learn library was chosen for this task. Random Forests are an ensemble of decision trees trained on bootstrap samples of the data with random feature subsets at each split (bagging). This approach reduces variance while maintaining low bias, making it well-suited for tabular datasets with mixed feature types.

The model was configured with the following hyperparameters:

• n_estimators = 200 (number of trees in the forest)

• max_depth = 10 (maximum depth per tree to prevent overfitting)

• random_state = 42 (for reproducibility)

3.4 Train-Test Split

The dataset was divided into training and test sets using an 80/20 split ratio. Stratification was not applied since Profit is a continuous variable. The random_state parameter was fixed at 42 to ensure reproducibility of results.

4. Implementation

4.1 Technology Stack

• Python — Primary programming language

• pandas — Data loading, manipulation, and feature engineering

• numpy — Numerical operations

• scikit-learn — Model training (RandomForestRegressor), metrics, and train-test split • matplotlib — Visualisations (scatter plot and bar chart)

• Jupyter Notebook — Interactive development environment

4.2 Pipeline Overview

The implementation followed a structured ML pipeline:

• Step 1 — Data Ingestion: Load Excel file into a pandas DataFrame.

• Step 2 — Date Parsing: Convert string date columns to datetime.

• Step 3 — Feature Engineering: Derive temporal features and shipping duration. •

Step 4 — Column Cleanup: Drop identifier columns.

• Step 5 — Encoding: One-hot encode all categorical features.

• Step 6 — Target Split: Separate Profit as the target variable (y), all other columns as features (X).

• Step 7 — Train-Test Split: 80% train, 20% test.

• Step 8 — Model Training: Fit RandomForestRegressor on training data. •

Step 9 — Prediction: Generate predictions on the test set.

• Step 10 — Evaluation: Compute R^2, MAE, and RMSE.

• Step 11 — Visualisations: Actual vs. Predicted scatter plot and Top 15 Feature Importances bar chart.

5. Results and Analysis

5.1 Model Performance Metrics

Three metrics were used to evaluate the performance of the Random Forest model: R^2 Score; (MAE) Mean Absolute Error; (RMSE) Root Mean Square Error;.

The model got a (R^2) value of 0.8752. This (R^2) value means that about 87.52% of the profit value used in this dataset can be explained by this Random Forest model. (90% is generally accepted to be strong/acceptable/excellent predictive capabilities for any retail datasets. 87.52% is an excellent prediction strength for this retail dataset).

With respect to accuracy of the predictions the (MAE) for the Random Forest model was 14.23 or an average of $14.23 of actual versus predicted profit. Given the number of transactions in this retail dataset and the inherent variability of retail transaction level profit, this is a reasonable error rate for the insights provided by this Random Forest model.

The RMSE of the Random Forest model was 38.71 and is higher than the MAE. There are a few predictions made by the Random Forest model that are significantly less accurate and therefore extremely larger than the average in both error values, and they tend to occur only when a retailer has significantly larger than average discounts and/or has some unusually large orders.

Overall, the Random Forest model has performed very well and has adequately captured the underlying retail profit patterns for this retail dataset.

5.2 Actual vs. Predicted Profit

The scatterplot of Actual and Predicted Profits shows a high degree of linearity, which means that the model has been able to capture the pattern of profit quite accurately in this range. Most predicted values lie close to the diagonal identity line. However, some variation exists in the outliers, especially for highly positive values of profits (> $200) and negative values (losses).

5.3 Feature Importance Analysis

The Random Forest algorithm gives a feature importance rank for determining profit. Therefore, we can analyze how each variable influences profit in our study.

From the analysis, it was evident that sales was the top predictor of profit among all the features considered. It is expected since the higher the sales, the more profit will be generated assuming that discount levels are kept at low levels.

The next predictor of profit was the discount variable, which had a high negative coefficient of importance. The higher the discount, the lower the profit will be generated. It implies that there should be strict control in managing the discount variable to maximize profit generation.

The quantity variable is important since it dictates the total number of items

sold. Moderately important predictors are:

- Number of shipping days – It dictates the efficiency of shipment processes and costs
- Order month – It captures the seasonality effect of demand for products in various months of the year
- Sub-category – It denotes what kind of product was sold

Variables such as region and segment are less influential but still important since they help improve the prediction accuracy of the model.

6. Key Business Insights

Beyond model accuracy metrics, the analysis yields several actionable insights for retail management:

Discount Rate is a Profit Killer

Feature importance analysis and exploratory data patterns confirm that orders with discount rates above 30-40% are disproportionately likely to result in negative profit. The model effectively learns this boundary. Recommendation: Implement discount caps by product sub-category, especially for Furniture (Tables, Bookcases) and Office Supplies where margin is thin.

Technology Sub-Category Drives the Highest Margins

Products in the Technology category, like phones, copiers, and accessories, always make the most money. The model gives sub-category features in Technology a lot of weight. Recommendation: Focus on upselling technology products and give fewer discounts in this category because the demand is not very elastic.

The South and Central Regions are Less Profitable

The region characteristics help formulate predictions from the model, whereby the Southern and Central regions show tendencies towards lesser profit margins than the Eastern and Western regions. This could be attributed to logistics expenses, different prices, or competition. Recommendation: Evaluate the cost structure in the regions and adjust pricing/discounting policies.

Shipping Mode Matters

The engineered Shipping Days feature captures the effect of shipping mode on profit. Same Day and First Class deliveries incur higher shipping costs, reducing net profit per order. Recommendation: Encourage Standard Class shipping for low-margin orders and reserve expedited shipping for high-value Technology orders.

Seasonal Demand Patterns

The Order Month emerged as a relevant characteristic, as months falling in Quarter 4 (October December) witnessed higher orders associated with promotions at year-end. Nevertheless, promotion discounts in those months reduce revenues to some extent. Recommendation: Analyze profitability in relation to seasonal promotions prior to implementation.

7. Conclusion

Using the Sample Superstore dataset, a successful machine learning pipeline was built to estimate profit by order in retail through the use of a Random Forest Regressor created through the combination of the transactional features and engineered temporal/shipping features. The predictive accuracy of the model was strong with an R^2 of approximately 0.875.

The model demonstrates both technical proficiency in data preprocessing, feature engineering, and ensemble learning, as well as delivering business value through interpretable feature importances that indicate the key drivers of profit. Additionally, the strong impact of sales and discount feature relations to profit demonstrate established principles of retail management, while the importance of extracted features such as shipping days further supports the need for feature extraction based on domain knowledge.

Future improvements would include hyperparameter tuning through cross-validated grid search, experimentation with gradient boosting algorithms (XGBoost/LightGBM) to provide a comparative analysis, the addition of customer level aggregated features (e.g., lifetime value of customer), and the model can be deployed as a web API to provide profit estimates during

order processing.

8. References

• Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32. • Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

• McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.

• Tableau. Sample Superstore Dataset. Tableau Public Dataset Repository. • Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.

• Hunter, J.D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90–95.